

Supplemental information

Sexual dimorphism in the complete *Drosophila*

male central nervous system connectome

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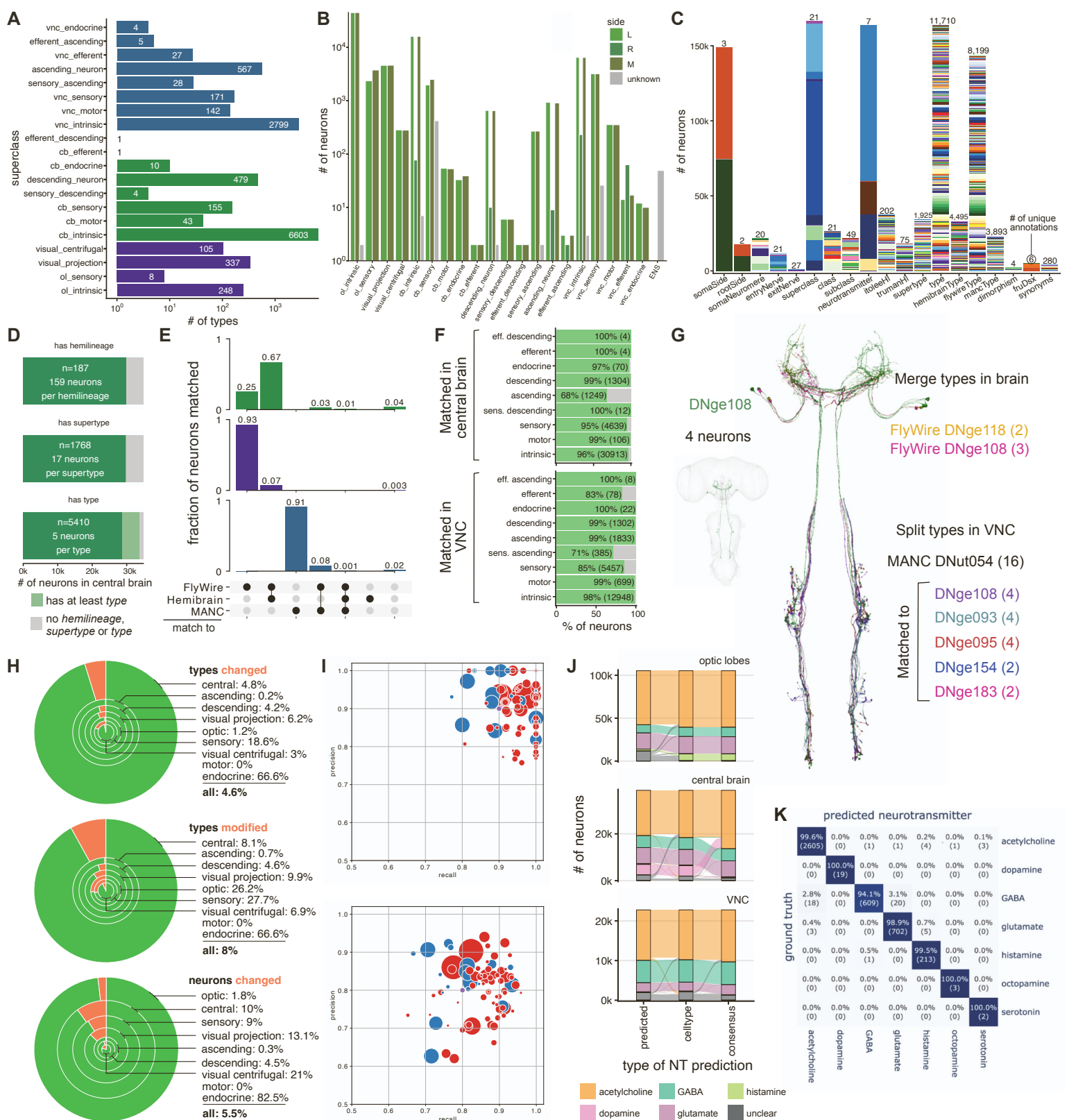


Figure S1 | Annotation, cross-matching and synapse and neurotransmitter assignment, related to figure 1 and figure S8

A Number of types per superclass and per CNS region. **B** Number of neurons by soma or root side per superclass. **B** Number of neurons with annotations for several categories. The number of unique annotations for each one is shown at the top of the bar. **D** Number of neurons in the central brain with *ito*leeH1, supertype and/or type annotations. The number of distinct annotations, and the average number of neuron per category is shown in the bar. **E** Fraction of neurons cross-matched to existing datasets per CNS region. **F** Number of cross-matched neurons in the central brain and VNC, per superclass. The number of matched neurons and the corresponding percentage is shown. **G** The descending neuron type DNge108 was matched to 2 types in FAFB/FlyWire and a subset of neurons for type DNut054 in MANC. The remaining 12 neurons in DNut054 match 4 other types in male CNS. **H** FlyWire cell types updated through cross-matching with the male CNS, broken down by superclass. Top: percentage of types which were removed/replaced entirely; middle: percentage of types for which at least one neuron changed; bottom: percentage of neurons for which the type changed. Changes in type names due to adopting the Matsliah *et al.* (2024)¹ nomenclature for optic lobe-intrinsic neurons are not counted. Surface area of wedges corresponds to the total number of neurons/types in that category. **I** The precision and recall of synapse T-bar predictions (top) and pre-post connections (bottom) in each region of interest (ROI). Markers are scaled by ROI size and color differentiates between brain (red) and VNC (blue) neuropils. **J** Change in neurotransmitter prediction for predictedNt, celltypeNt and consensusNt for each neuron in the CNS subregions. **K** Neuron-level confusion matrix for neurotransmitter prediction (predictedNt) on the held-out test set.

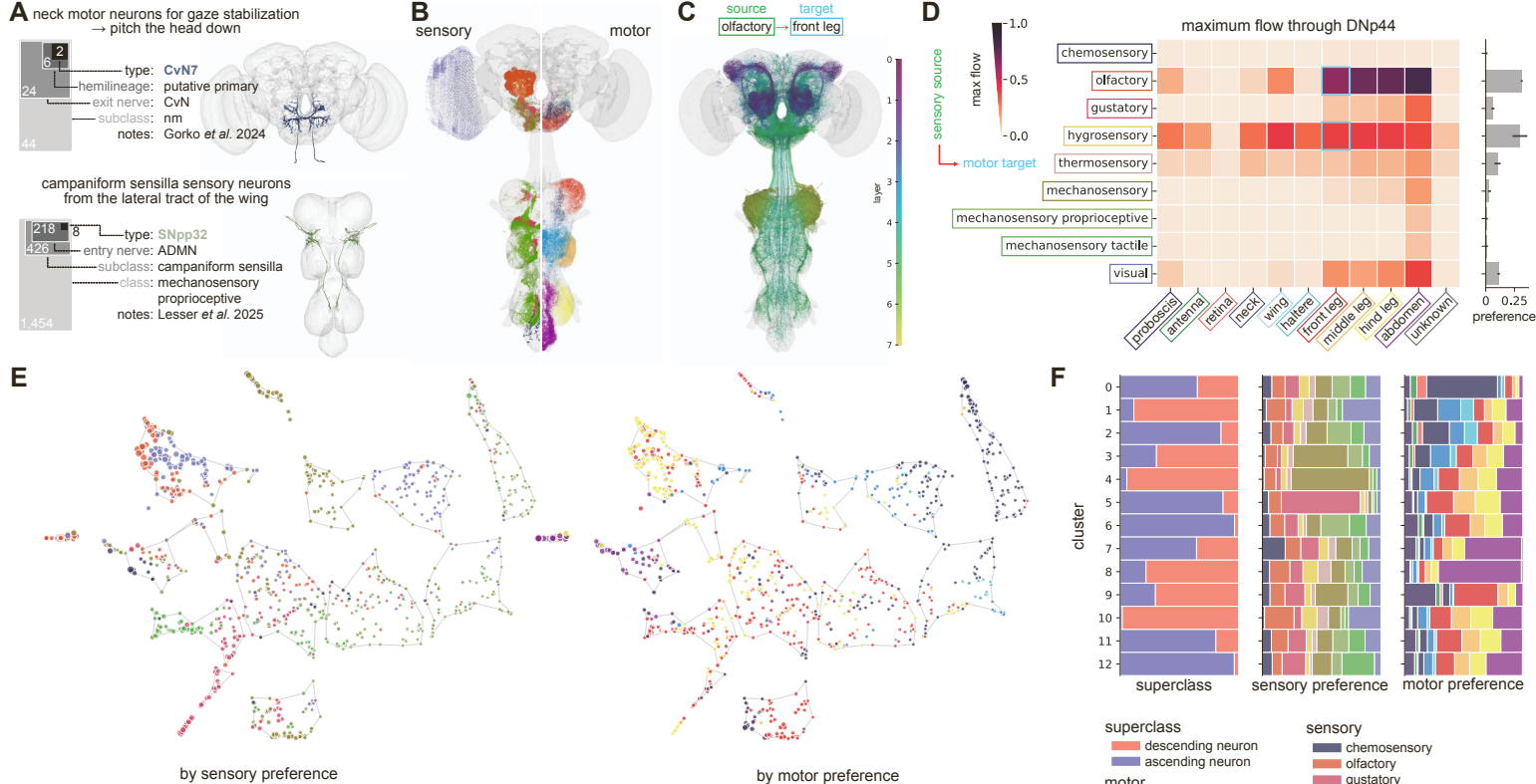


Figure S2 | Additional flow analysis, related to figure 2. **A** Sample annotation for sensory and motor neurons. Top: neck motor neuron CvN7. Bottom: wing campaniform sensilla sensory type SNpp32. Treemaps show how annotations can be intersected to filter neurons. **B** Synapses from sensory (left) or motor groups (right) in the CNS, colored by scheme in Figure 2A. **C** Example of maxflow routes from olfactory inputs to front leg motor outputs: neurons (olfactory flow value > 0.25) colored according to their pseudo-layer ordering. **D** Example of mean maxflow through descending neuron type DNp44 (heatmap) and sensory modality preference averaged from flow (histogram). Highlighted squares correspond to flows shown in Figure 2D. **E** UMAP embedding of DN and AN types from Figure 2E colored by preferred sensory modality (left) and motor (right), sized by maximum preference value. **F** Composition of superclass, sensory preference and motor preference for clusters in Figure 2E and 2F.

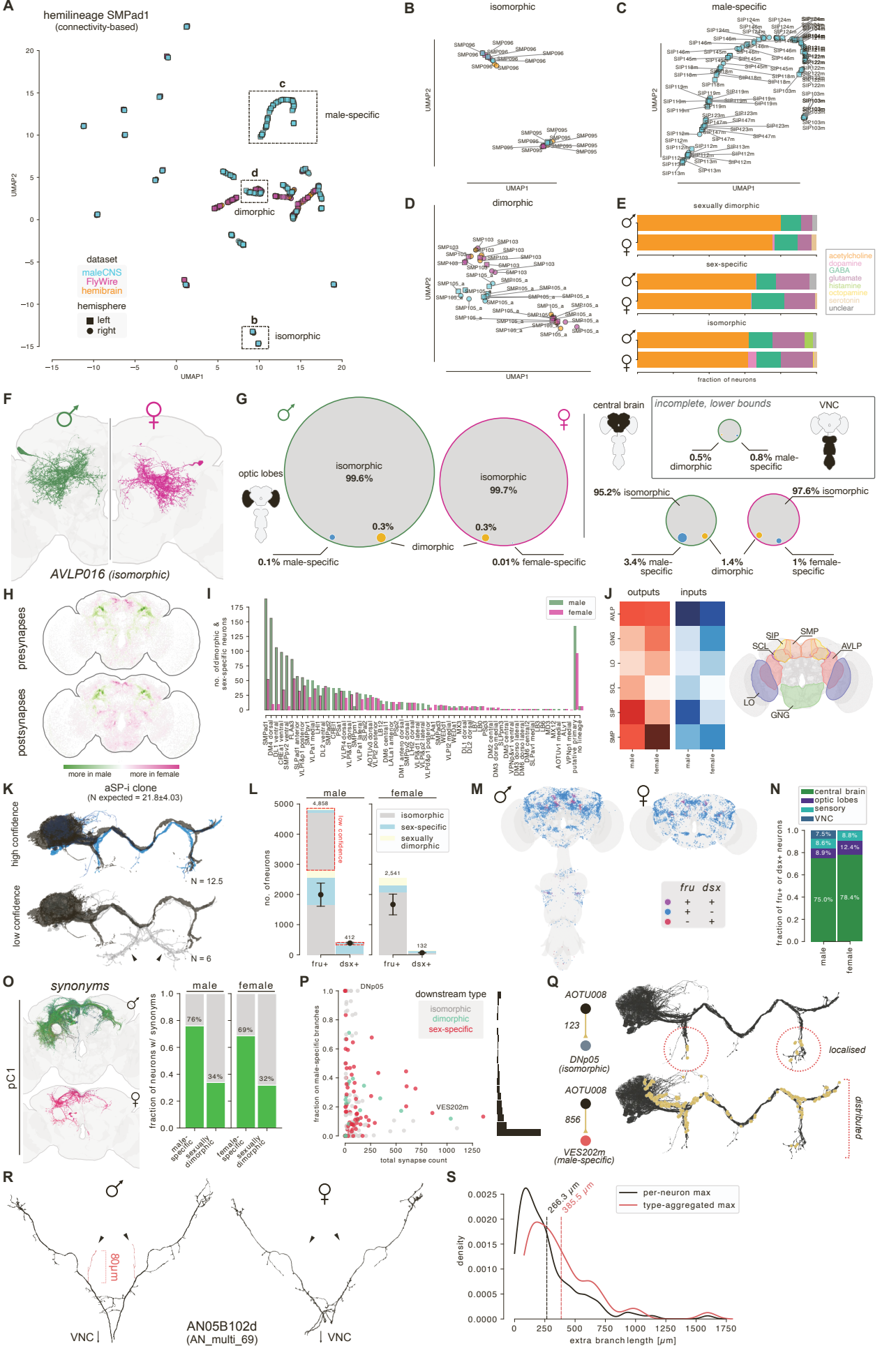


Figure S3 | Additional dimorphism analysis, related to figure 3

A UMAP embedding of connectivity-based cosine distances for neurons of the SMPad1 hemilineage from three connectomes/five hemispheres: male CNS, FAFB/FlyWire (female) and hemibrain (female). **B-D** Zoom-ins on insets labeled in a showing isomorphic (b), male-specific (c) and dimorphic (d) cell types. **E** Distribution of predicted neurotransmitters for dimorphic, sex-specific and isomorphic neurons. **F** Male and female version of AVLP16. **G** Pie charts showing the fraction of dimorphic and sex-specific neurons across optic lobes, central brain and VNC. Relative sizes correspond to the number of neurons. For the VNC, our annotations in the male include 15 dimorphic types (47 neurons) from previous publications²⁻¹² that are either VNC-intrinsic (i.e. do not have a correlate within the female brain volume), or ascending neurons for which the axonal arbors in the brain are too small to identify them in FAFB/FlyWire. Note that VNC dimorphism labeling is partial and hence represents a lower bound. **H** Difference between distribution of pre- and postsynapses from dimorphic/sex-specific neurons in males and females. **I** Number of dimorphic and sex-specific neurons per hemilineage in male (green bars) and female (purple bars). Lineages with bars outlined in black collectively produce ≥50% of dimorphic and sex-specific neurons in that sex. **J** Top 6 neurolins with the most dimorphic/sex-specific synapses. **K** Example for high- and low-confidence assignment of *fruitless* (*fru*+) annotation. **L** Breakdown of *fruitless* (*fru*+) and *doublesex* (*dsx*+) annotations by dimorphism. This represents an upper boundary because most of the light-level data labels large populations of neurons which makes identifying individual neurons difficult. Dotted red outline indicates low-confidence annotations in the male. Pointplots in black represent mean and standard deviation of expected counts based on prior literature^{5,9}. **M** Cell bodies of *fru*+/ and *dsx*+/ neurons in male (left) and female (right). A small number of *fruitless* and *doublesex*-expressing neurons were annotated in the VNC based primarily on previous annotations from the MANC dataset^{3,13}. **N** Distribution of *fru*+/ and/or *dsx*+/ neurons across main brain regions. The majority (75%) of *fruitless/doublesex* annotations are in the central brain, 7.5% are in the VNC, and 8.9% are in the optic lobes. **O** Left: pC1 as an example for a synonym assigned to a group of dimorphic neurons. Right: fraction of dimorphic neurons matched to prior literature. **P** For each outgoing connection of AOTU008, the fraction of synapses from male-specific branches versus total synapse count. **Q** Example for distributed connections and connections highly localised to male-specific branches of AOTU008 in the male. **R** Ascending dimorphic cell type AN05B102d (AN_multi_69 in FlyWire) has one of the smallest sex-specific branches (80μm). Arrowheads indicate male-specific branches (red). Only the brain-portion of the neurons shown. **S** KDE (kernel density estimate) of the distribution of male-specific branch lengths: largest male-specific branch per neuron (black) and aggregated per cell type (red).

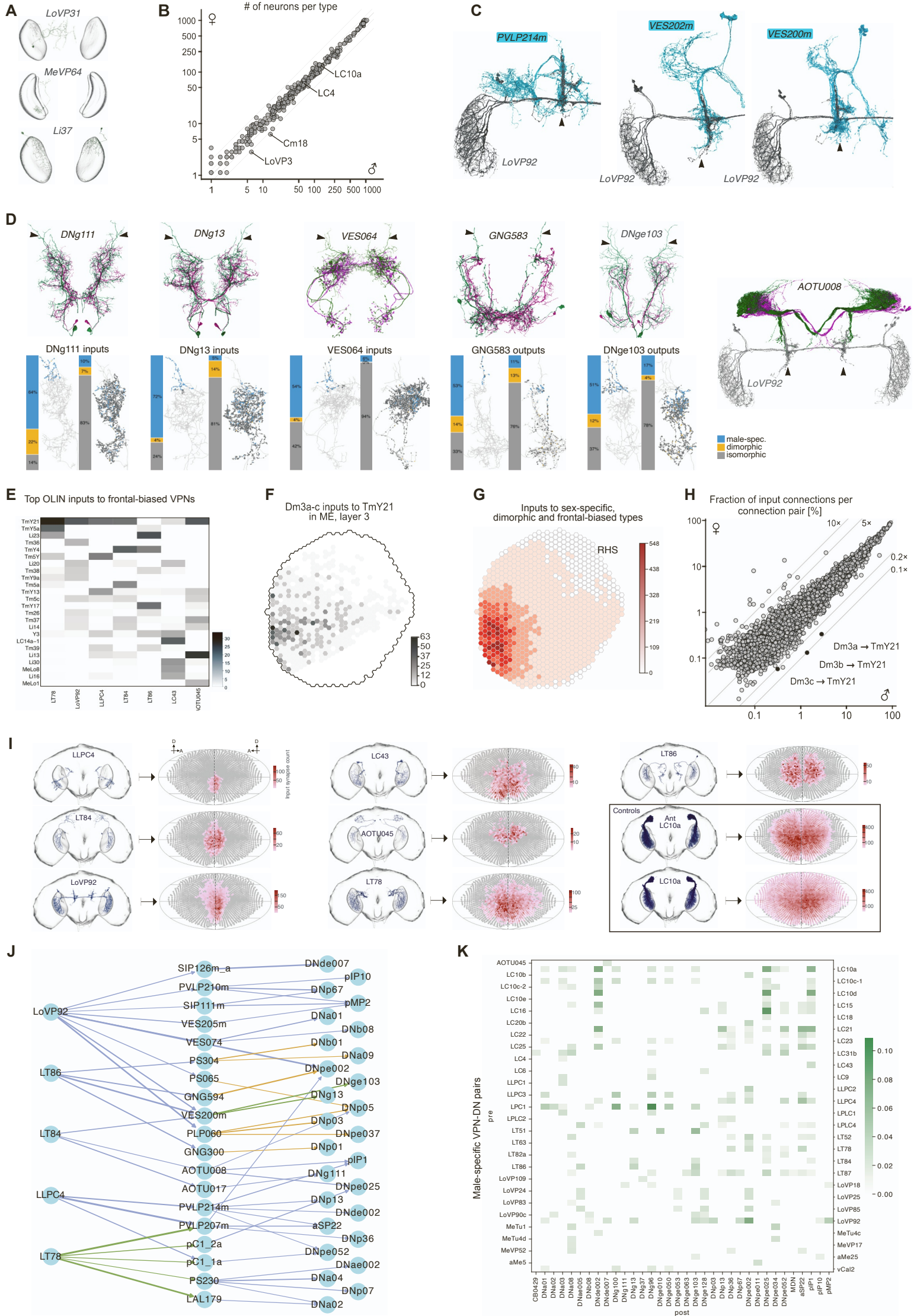


Figure S4 | Additional visual system analysis, related to figure 4

A Unmatched outlier types were excluded from classification as male-specific due to a lack of left-right symmetry. Note that *Tm40* is currently classified as isomorphic. However, FAFB/FlyWire has a lower cell count compared to the male CNS (41 versus 180 cells). Consequently, it was excluded from the analysis shown in B and H. **B** Male versus female neuron counts per type (one-to-one matches only). Female counts were normalised to account for ~100 additional male ommatidia. Photoreceptors and *Lai* are excluded. **C** Male-specific and **D** dimorphic types (male shown in green, female in magenta) converging on the male-specific node defined by *LoVP92* axons (arrowheads). Stacked bar charts break down the synaptic input onto male-specific (left) versus isomorphic (right) arbour based on dimorphism status of upstream partner neurons. **E** Heatmap of the strongest OLIN inputs to frontal-biased VPNS. Scale bar shows % input. **F** Spatial map of *Dm3a-c* inputs to *TmY21* (ME, layer 3). **G** Inputs to male-specific, dimorphic and secondary dimorphic frontal types are biased towards columns devoted to the frontal field of view. **H** Input-normalized synaptic strength between pairs of OLIN types in male versus female. Only edges ≥ 500 synapses shown. Envelopes indicate 5- and 10-fold changes in relative weight. **I** Morphology (left) and input synapse distribution mapped onto a Mollweide projection of the visual field for both compound eyes¹⁴ (right) for example frontal-biased VPNS. **J** All pathways from VPNS in Figure 4H, to DNs with known functions or sexual dimorphism, within two synaptic hops, where all connections are stronger than 1% input. **K** Effective connectivity based on male specific pathways, involving at least one dimorphic or sex-specific cell type, within two synaptic hops, where each connection is stronger than 1% input, from all VPNS, to DNs with known functions or sexual dimorphism. The colour is based on the sum of direct connection strength, and the square root of the products of weights in the two-hop pathways.

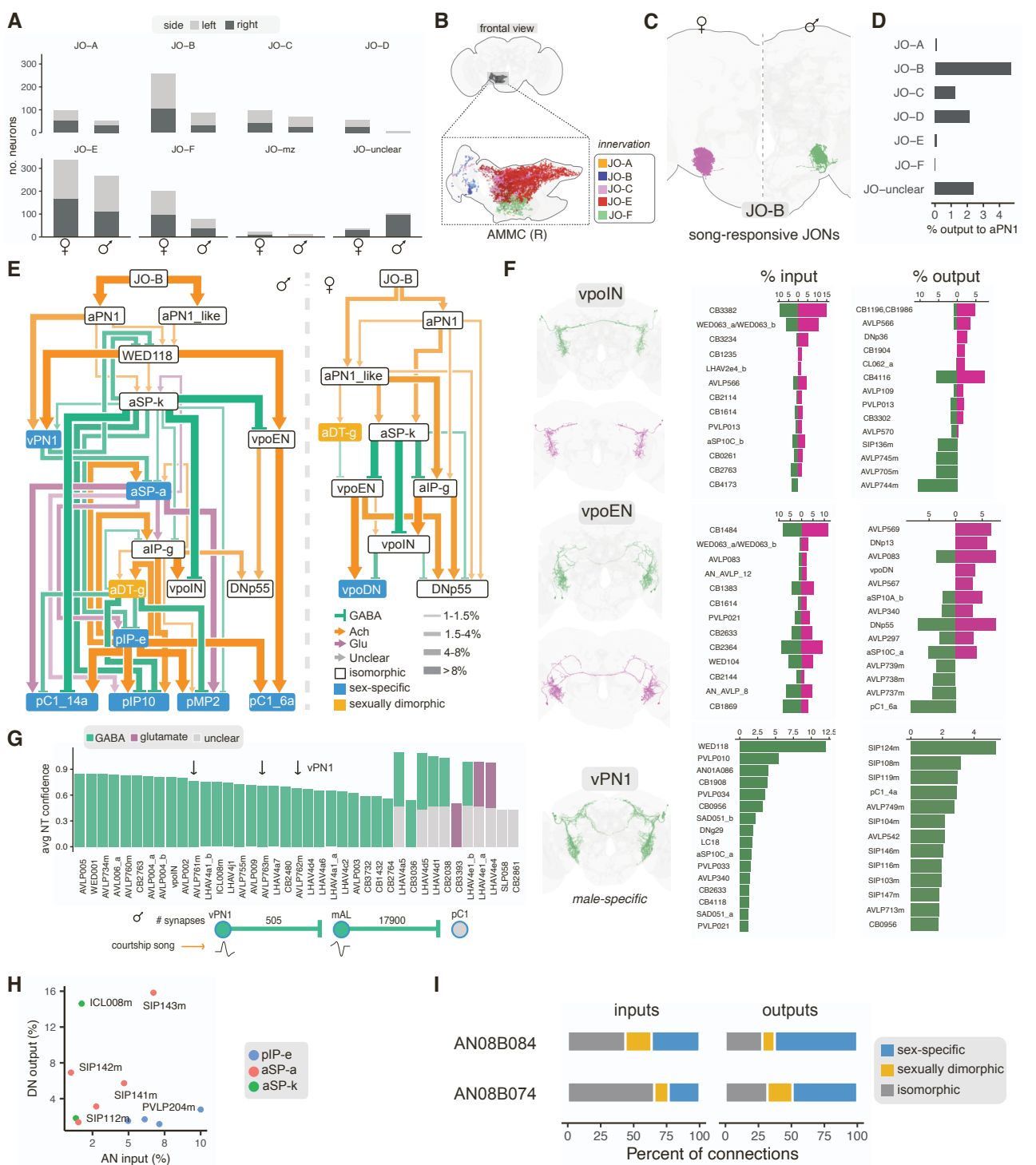


Figure S5 | Additional auditory system analysis, related to figure 5

A Johnston's Organ neuron count by type in males (male CNS) and females (FAFB/FlyWire). Despite segmentation issues, we were able to identify around 20% of JO-B neurons in male CNS. **B** AMMC innervation by JO neuron types in male CNS. **C** JO-B neurons in female (FAFB/FlyWire, magenta) and male (male CNS, green) brains. **D** Proportion of outputs from JO neuron types to aPN1. **E** Circuit schematics for male (left) and female (right) from JO-B to higher order processing centres. Boxes denote cell types, edge thickness scales with synaptic weight, and edge colors indicate transmitter identity. Node colors mark sex classes: isomorphic (white), sexually dimorphic (gold), sex specific (blue). Line widths denote input-normalized connection strengths. **F** Top input (left) and output (right) partners of vpoIN, vpoEN, and vPN1 in males (green) and females (magenta). Bars show the fraction of total synaptic inputs (input-normalized) and outputs (output-normalized) to each identified type across sexes. **G** Average confidence score for consensus neurotransmitter predictions in the LHI1 lineage in male CNS; vPN1 cell types indicated by black arrows. Neuron schematic illustrates disinhibition of pC1 by vPN1 via mAL interneurons. **H** Scatter plot of the synaptic connection strength with song detection (y-axis) and AN feedback (x-axis) for cell types comprising *fruitless*-expressing clones in the circuit diagram in Figure 5F. **I** Fraction of sexually dimorphic (yellow) and sex specific (blue) inputs and outputs for AN08B074 and AN08B084.

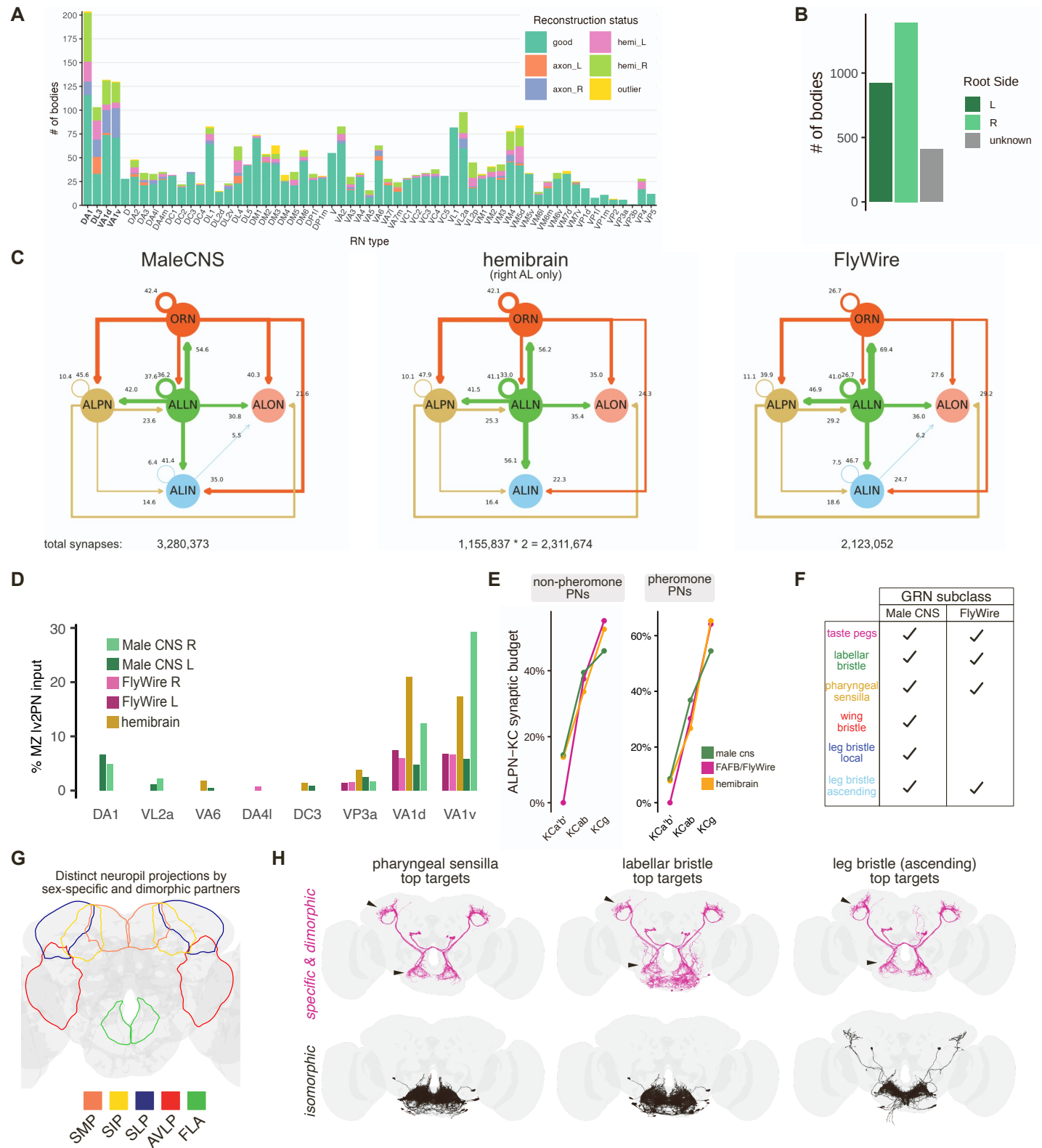


Figure S6 | Proofreading and connectivity analysis in the chemosensory systems, related to figure 6

A Number of bodies per RN (O/T/HRN) type and its reconstruction status. axon_L/R: axon (presynapses) only present on that side, though the body crosses the midline. hemi_R/L: body ends around the midline, and only exists on one side. Outlier: cases in which a body innervates more than one glomerulus on one side or does not arborize a glomerulus proper. **B** Number of bodies per RN type and per root side. **C** Input-normalized wiring diagram between AL cell classes for AL-intrinsic connectivity in males (left) and females (FAFB/FlyWire, right). **D** Barchart of top RN input (over 0.1% per type) to MZ_lv2PN in male and female brains. **E** Comparison of the synaptic budget, as percentage for the connections from ALPNs for the main types of Kenyon cells in the male and female brains; the differences observed between datasets for the KC classes are possibly due to biological variability. Left: non-pheromone ALPNs; Right: pheromone ALPNs. **F** Left: Table comparing male CNS subclasses of taste sensory neurons with FAFB/FlyWire data. Wing bristles and local leg bristles are exclusive to the VNC. **G** Strongest neuropils innervated by second-order sex-specific and sexually dimorphic gustatory neurons (SMP, superior medial protocerebrum; SIP, superior intermediate protocerebrum; SLP, superior lateral protocerebrum; AVLP, anterior ventrolateral protocerebrum; FLA, flange). **H** Top ten targets downstream of GRN types in female (FAFB/FlyWire), by type.

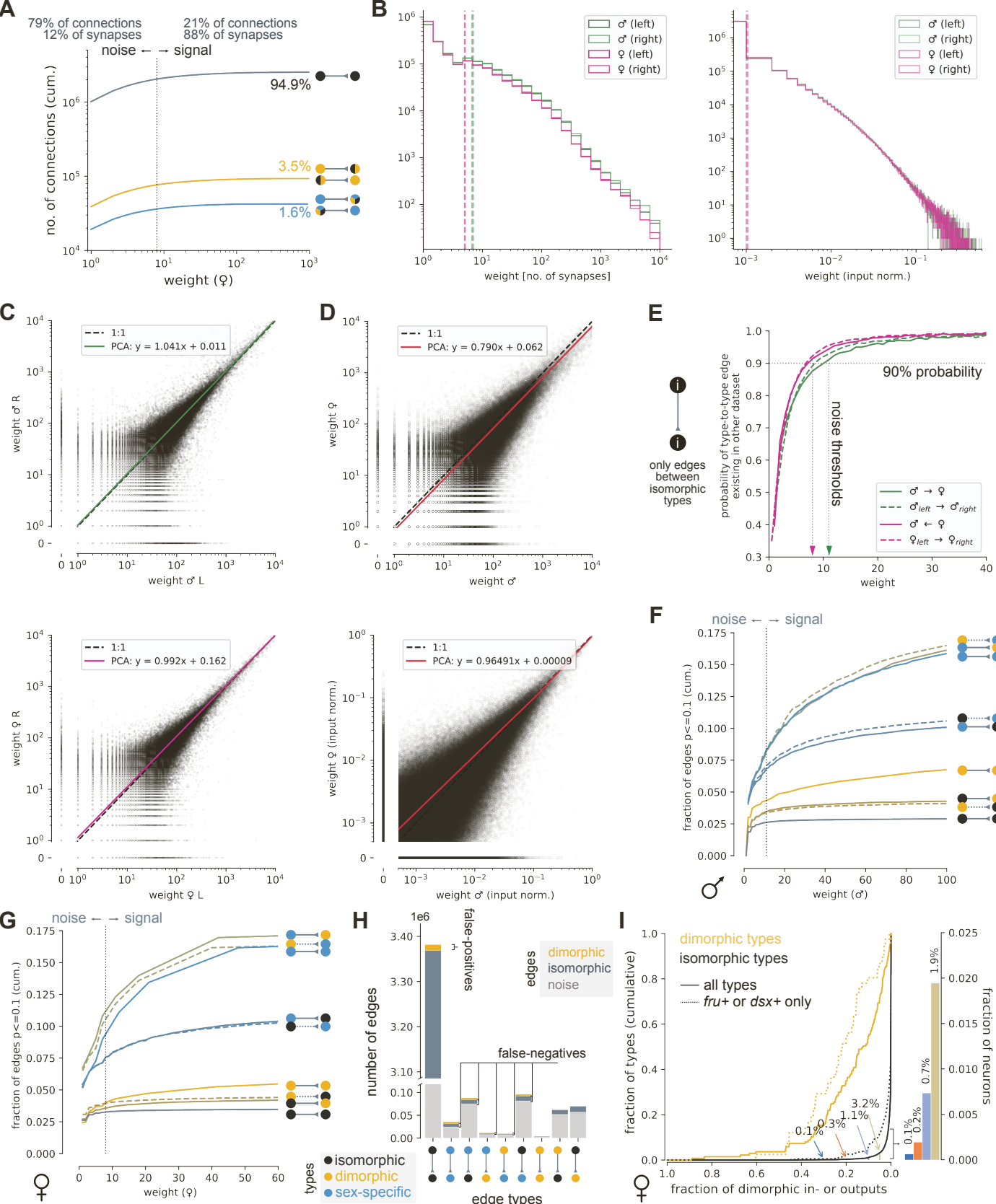


Figure S7 | Supporting dimorphism analyses, related to figure 7

A Distribution of female type-to-type connection weights broken down by dimorphism of pre- and postsynaptic cell type. Dotted vertical line indicates noise threshold. **B** Histogram of total (left) and input-normalised (right) cross-matched type-to-type edge weights for both male and female. Dotted vertical lines indicate the means. **C** Left versus right weights for cross-matched type-to-type edges in male (top) and female (bottom). Green/magenta line represents PCA fit. **D** Male versus female edge weights for cross-matched types. Top: total number of synapses; bottom: input normalised. Red lines represent PCA fit. **E** Probability of finding an edge between isomorphic neurons of a given weight in another hemisphere of the same or different dataset. **F** Cumulative fraction of male edges with FDR-corrected p-values below 0.1 per edge type. Dotted vertical line marks 90% probability threshold from E. **G** Same as F but in female. **H** Total number of dimorphic, isomorphic and noisy connections (male + female) broken down by edge type. **I** Cumulative fraction of iso- and dimorphic cell types (y axis) with more than a given fraction inputs or outputs (x axis, by synapse count) in female. Arrows indicate fractions at various thresholds. For example, 0.1% of types have $\geq 30\%$ dimorphic in- or outputs. Bar chart on the right translates thresholds into fraction of neurons.

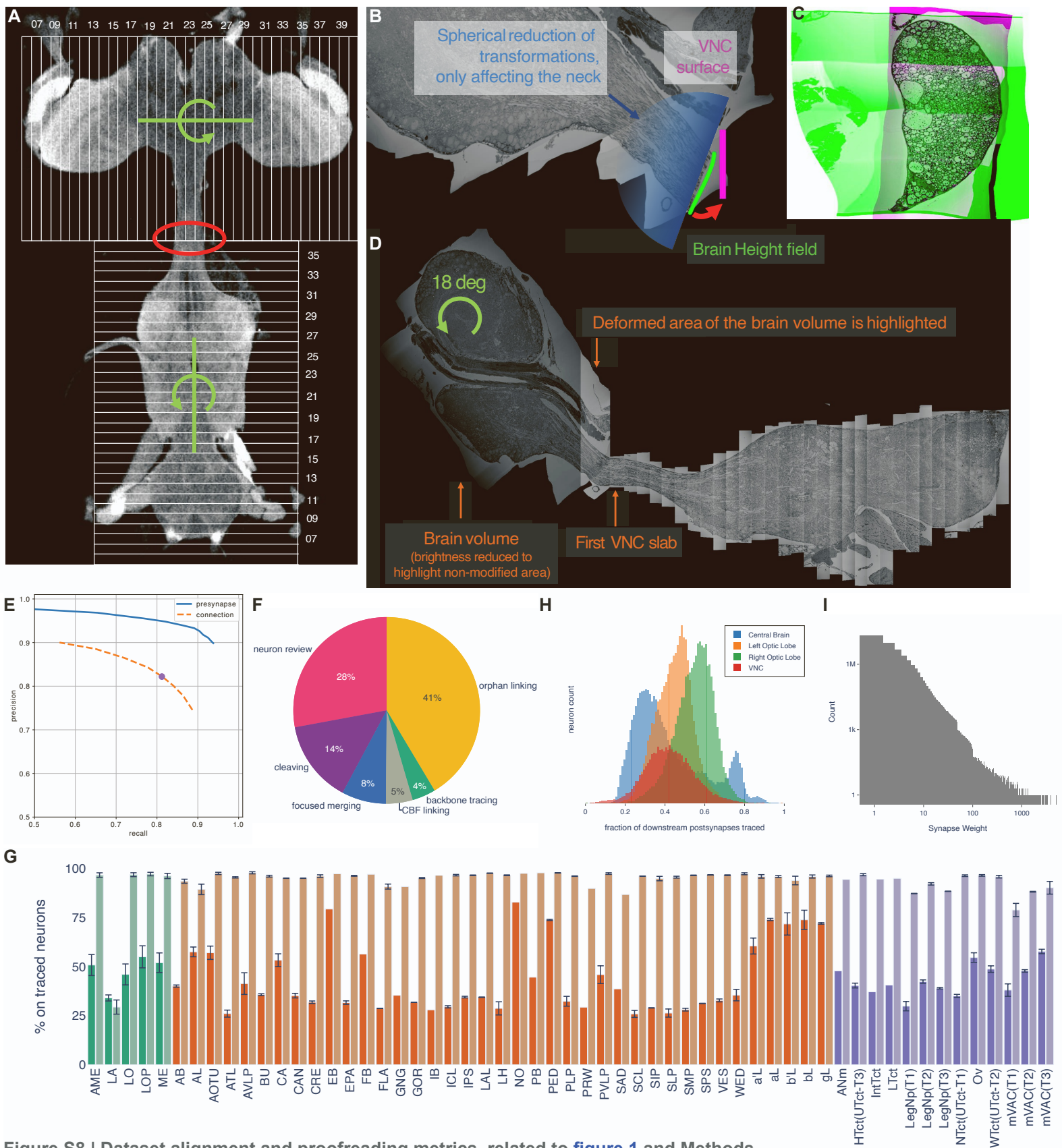


Figure S8 | Dataset alignment and proofreading metrics, related to figure 1 and Methods

A Overview of the entire CNS and the slicing into slabs for eFIB-SEM imaging. The red ellipse highlights the brain/VNC cut, green rotation axes show the degrees of freedom wrt to slab orientation. **B** Transformations that were applied to the brain and VNC volumes to stitch them seamlessly. **C** Overlay of the extracted surfaces of the brain (green) and VNC (magenta). Note the 4 slabs that are visible in the brain surface. **D** Cut through the entire CNS volume highlighting the connection between brain and CNS, the modified area of the brain for stitching both volumes, and the slabs of the VNC. **E** The precision-recall curves for presynapses and connections, sweeping across the model's output confidence score. Our published synapses are filtered with a confidence threshold of 0.5 (purple dot), resulting in overall precision of 0.82 and recall of 0.81. **F** Estimated proportion of proofreading labor spent in each major proofreading protocol. **G** Traced completeness of synapses across neuropils. Light bars show presynapse completeness, dark bars show postsynapse completeness, and hue differentiates between optic lobe, central brain, and VNC neuropils. For paired left and right neuropils, the whiskers indicate the spread between left and right values and the bar shows the mean. **H** The distributions of downstream capture fraction for traced neurons in the central brain, optic lobes, and ventral nerve cord. **I** The distribution of synaptic partner counts of remaining unmerged 'orphan' fragments.

Supplemental references

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